

Notes: Tamer (RES, 2003)

Incomplete Simultaneous Discrete Response Model with Multiple Equilibria

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1 Big picture

- **Question.** Can a bivariate simultaneous discrete-response model with multiple equilibria still identify structural parameters and be estimated without imposing ad hoc selection rules or “coherency” restrictions that kill simultaneity?
- **Setting.** The paper studies a two-player binary game whose equilibrium outcomes generate a standard simultaneous discrete-choice econometric model. Multiplicity in the underlying game shows up as regions of the covariate space where the econometric model predicts more than one feasible outcome.
- **Core challenge.** In the usual reduced-form interpretation, the model can look “incoherent” because probabilities assigned to the four outcomes can sum to more than one. Earlier papers often fixed this by imposing coherency conditions or by arbitrarily selecting one equilibrium in the multiplicity region.
- **Main conceptual contribution.** Tamer argues that this is better viewed as an **incomplete econometric model**, not necessarily a misspecified or useless one. Multiplicity delivers **inequality restrictions** on outcome probabilities rather than a standard full likelihood.
- **Main identification result (Theorem 1).** When the interaction parameters Δ_1 and Δ_2 have the same sign, the parameter vector $\beta = (\beta_1, \beta_2, \Delta_1, \Delta_2)$ is point identified under support and exclusion conditions, even though some outcomes are only bounded, not exactly predicted.
- **Main identification result (Theorem 2).** When $\Delta_1 \Delta_2 < 0$, so that the model provides only upper and lower bounds for every outcome, the parameter vector is still point identified if at least one continuous excluded regressor has sufficiently rich support.
- **Main estimation contribution.** In the same-sign case, the paper proposes a semiparametric maximum-likelihood estimator (SML) that keeps the exact probabilities where the model is sharp and replaces the nonunique probability $\Pr[(0, 1) \mid x]$ with a nonparametric estimate trimmed to lie inside the model-implied bounds.
- **Main efficiency result.** Exploiting multiplicity is not just legitimate; it is useful. The SML estimator is more efficient than the limited maximum-likelihood estimator that collapses $(0, 1)$ and $(1, 0)$ into one combined event.
- **Main economic intuition.** Multiple equilibria do not destroy information. They create a region where the data reveal inequality restrictions, and those restrictions can be informative enough for point identification and more efficient estimation.
- **Takeaway.** The right response to multiple equilibria is not automatically to impose recursivity or arbitrary equilibrium selection. Incomplete models can still be point identified, estimable, and economically meaningful.

2 Model environment and key objects

2.1 The structural game and the econometric mapping

The paper studies the simultaneous latent-index system

$$y_1^* = x_1\beta_1 + y_2\Delta_1 + u_1, \quad (1)$$

$$y_2^* = x_2\beta_2 + y_1\Delta_2 + u_2, \quad (2)$$

with observed binary outcomes

$$y_j = \begin{cases} 1 & \text{if } y_j^* \geq 0, \\ 0 & \text{otherwise,} \end{cases} \quad j = 1, 2. \quad (3)$$

Here:

- $x = (x_1, x_2)$ is a vector of observed exogenous variables,
- $u = (u_1, u_2)$ is a latent payoff shock with joint distribution F_u ,
- $\beta = (\beta_1, \beta_2, \Delta_1, \Delta_2)$ is the vector of structural parameters.

Interpretation. This is the standard simultaneous binary-response model used in entry, social interaction, and labor-supply applications. The novelty in the paper is not the structural system itself, but what to do when it generates multiple equilibria.

2.2 Coherency versus incompleteness

If Δ_1 and Δ_2 are both negative, the probabilities generated by the four best-response regions satisfy

$$\Pr[(0, 0) \mid x] = \Pr(u_1 < -x_1\beta_1, u_2 < -x_2\beta_2), \quad (4)$$

$$\Pr[(1, 1) \mid x] = \Pr(u_1 \geq -x_1\beta_1 - \Delta_1, u_2 \geq -x_2\beta_2 - \Delta_2), \quad (5)$$

$$\Pr[(0, 1) \mid x] = \Pr(u_1 < -x_1\beta_1 - \Delta_1, u_2 \geq -x_2\beta_2), \quad (6)$$

$$\Pr[(1, 0) \mid x] = \Pr(u_1 \geq -x_1\beta_1, u_2 < -x_2\beta_2 - \Delta_2). \quad (7)$$

In that case,

$$\Pr[(0, 0) \mid x] + \Pr[(0, 1) \mid x] + \Pr[(1, 0) \mid x] + \Pr[(1, 1) \mid x] > 1. \quad (8)$$

Earlier work often imposed the **coherency condition** $\Delta_1\Delta_2 = 0$, which is necessary and sufficient for the four outcome probabilities to sum to one.

Interpretation. Tamer's central point is that this "incoherency" comes from interpreting an equilibrium correspondence as if it were a function. Once we recognize the model as **incomplete**, the multiplicity region becomes informative rather than pathological.

2.3 Outcome regions and incomplete probabilities when $\Delta_1, \Delta_2 < 0$

When both interaction terms are negative, the model gives **exact** probabilities for the two diagonal outcomes and **bounds** for the off-diagonal outcomes:

$$P_1(x, \beta) = \Pr[(0, 0) | x] = \Pr(u_1 < -x_1\beta_1, u_2 < -x_2\beta_2), \quad (9)$$

$$P_2(x, \beta) = \Pr[(1, 1) | x] = \Pr(u_1 \geq -x_1\beta_1 - \Delta_1, u_2 \geq -x_2\beta_2 - \Delta_2), \quad (10)$$

$$P_3(x, \beta) \leq \Pr[(0, 1) | x] \leq P_4(x, \beta), \quad (11)$$

where

$$P_3(x, \beta) = \Pr(u_1 < -x_1\beta_1 - \Delta_1, u_2 > -x_2\beta_2 - \Delta_2) \\ + \Pr(u_1 < -x_1\beta_1, -x_2\beta_2 < u_2 < -x_2\beta_2 - \Delta_2), \quad (12)$$

$$P_4(x, \beta) = \Pr(u_1 < -x_1\beta_1 - \Delta_1, u_2 > -x_2\beta_2). \quad (13)$$

A symmetric pair of bounds holds for $\Pr[(1, 0) | x]$.

Interpretation. The model knows exactly how often the unambiguously low and unambiguously high outcomes occur. It is only in the middle region that the data tell us an interval rather than a point probability.

2.4 Why incompleteness can still identify parameters

The paper adopts Manski's distinction:

- a **complete** model maps (x, u) into a unique y ;
- an **incomplete** model maps (x, u) into a correspondence of feasible y 's.

Interpretation. Incompleteness and identification are logically distinct. A model can fail to deliver a unique conditional distribution for outcomes and still pin down the parameter vector uniquely through the inequalities it implies.

2.5 Example 1: Cournot entry game

The paper shows that a standard two-firm Cournot entry model maps into the same econometric structure. With inverse demand $p = a - bq$, firm payoffs imply

$$\Delta_1 = \Delta_2 = -\frac{5a^2}{36b}. \quad (14)$$

The game can have two asymmetric equilibria:

$$(\text{Enter}, \text{No entry}) \quad \text{and} \quad (\text{No entry}, \text{Enter}).$$

Interpretation. This example is important because it shows that the incomplete econometric model is not an abstract pathology. It naturally arises in empirical entry models where multiple equilibria are the object of interest.

2.6 Example 2: complementary technologies

In the second example, two agents supply complementary goods. The only Nash equilibria are $(1, 1)$ and $(0, 0)$. If $\theta_1 = \theta_2 = \theta < 1$, then in the multiplicity region the model implies

$$0 \leq \Pr(y = 1) \leq \theta^2. \quad (15)$$

Interpretation. Even when point identification fails, the incomplete model can deliver nontrivial and informative bounds. This example motivates the broader set-identification perspective that Tamer later develops with Manski.

2.7 Sampling and distributional assumptions

For the main identification and likelihood results, the paper assumes:

- an i.i.d. sample $\{(Y_{1i}, Y_{2i}), X_{1i}, X_{2i}\}$,
- $0 < \Pr[(Y_1, Y_2) \mid X_1, X_2] < 1$ for all outcomes,
- $U = (u_1, u_2)$ independent of X ,
- a known joint distribution F_u up to a finite-dimensional covariance matrix Ω ,
- and unit variances for u_1 and u_2 as a threshold-crossing normalization.

Interpretation. The paper uses these assumptions to get clean identification and asymptotic theory. It also notes that some of them, especially the known distributional form, are stronger than what is strictly required for identification alone.

3 Model and identification/estimation (all equations + interpretation)

3.1 Theorem 1: point identification when $\Delta_1 \Delta_2 > 0$

For the same-sign case, the paper focuses first on $\Delta_1 < 0$ and $\Delta_2 < 0$. Under the support and exclusion assumptions, $\beta = (\beta_1, \beta_2, \Delta_1, \Delta_2)$ is point identified.

The crucial support condition is: for at least one equation, there exists a regressor X_{ik} such that

- $\beta_{ik} \neq 0$,
- X_{ik} is excluded from the other equation,
- and $X_{ik} \mid X_{-ik}$ has an everywhere-positive Lebesgue density.

Interpretation. Exact probabilities for $(0, 0)$ and $(1, 1)$ allow the researcher to move one equation's index far enough in the tails to separate candidate parameter vectors. Once β_1 and β_2 are pinned down, the shift parameters Δ_1 and Δ_2 are recovered from the diagonal outcome probabilities.

3.2 Identification intuition in the same-sign case

The proof uses the fact that if $b \neq \beta$, then rich enough support in an excluded continuous regressor lets the researcher find x 's for which the exact probability under β differs from the exact probability under b . Symbolically, the exact probabilities

$$P_1(x, \beta) \quad \text{and} \quad P_2(x, \beta)$$

cannot be matched everywhere by the wrong parameter vector b .

Interpretation. Multiplicity does not contaminate every outcome. The exact-probability regions are strong enough to discipline the full parameter vector.

3.3 Sharper prediction under incompleteness

Once β is identified, the incomplete model provides tight bounds such as

$$P_3(x, \beta) \leq \Pr[(0, 1) \mid x] \leq P_4(x, \beta), \quad (16)$$

whereas a model that collapses $(0, 1)$ and $(1, 0)$ into a single event gives only the trivial bound

$$0 \leq \Pr[(0, 1) \mid x] \leq 1 - P_1(x, \beta) - P_2(x, \beta). \quad (17)$$

Interpretation. Keeping the incomplete structure is not only more faithful to the economics; it produces tighter empirical predictions.

3.4 Theorem 2: point identification when $\Delta_1 \Delta_2 < 0$

When $\Delta_1 > 0$ and $\Delta_2 < 0$ (the other case is symmetric), the model gives only upper and lower bounds for every outcome because the multiplicity region can support more than one feasible result. In particular, the paper defines

$$P_1(x, \beta) \leq \Pr[(1, 1) \mid x] \leq P_1(x, \beta) + P_{\square}(x, \beta), \quad (18)$$

$$P_2(x, \beta) \leq \Pr[(0, 0) \mid x] \leq P_2(x, \beta) + P_{\square}(x, \beta), \quad (19)$$

$$P'_3(x, \beta) \leq \Pr[(1, 0) \mid x] \leq P'_3(x, \beta) + P_{\square}(x, \beta), \quad (20)$$

where

$$P'_3(x, \beta) = \Pr(u_1 \geq -x_1\beta_1, u_2 < -x_2\beta_2 - \Delta_2), \quad (21)$$

$$P_{\square}(x, \beta) = \Pr(-x_1\beta_1 - \Delta_1 \leq u_1 \leq -x_1\beta_1, -x_2\beta_2 \leq u_2 \leq -x_2\beta_2 - \Delta_2). \quad (22)$$

Theorem 2 shows that β is still point identified if there exists at least one continuous excluded regressor with full support.

Interpretation. Even if every outcome is only interval-identified pointwise in x , the **parameter vector** can still be point identified once one exploits how the full system of inequalities moves with excluded continuous variation.

3.5 What to do when only bounds are available

In the opposite-sign case, the paper recommends a modified minimum-distance estimator in the spirit of Manski and Tamer (2002): choose the parameter vector that makes the model-implied inequalities fit the empirical conditional probabilities.

Interpretation. The message is practical: if the model is incomplete everywhere, do not force a false likelihood. Estimate the parameter from the inequality restrictions directly.

3.6 The limited maximum-likelihood estimator

For the same-sign case, the paper first reviews the common estimator that treats the two asymmetric outcomes as one joint event. The corresponding log-likelihood has the generic form

$$\mathcal{L}_{ML}(b) = \sum_i \left\{ 1[(0, 0)] \log P_1(x_i, b) + 1[(1, 1)] \log P_2(x_i, b) + 1[(0, 1) \text{ or } (1, 0)] \log (1 - P_1(x_i, b) - P_2(x_i, b)) \right\}. \quad (23)$$

Its asymptotic covariance matrix is

$$Q_{ML} = \left\{ E_x \left[\frac{\partial P_1}{\partial P_1} \frac{\partial P_1'}{\partial P_1} + \frac{\partial P_2}{\partial P_2} \frac{\partial P_2'}{\partial P_2} + \frac{(\partial P_1 + \partial P_2)(\partial P_1 + \partial P_2)'}{1 - P_1 - P_2} \right] \right\}^{-1}. \quad (24)$$

Interpretation. This estimator throws away the distinction between the two off-diagonal outcomes. It is valid, but it does not exploit all the information in the multiplicity region.

3.7 The semiparametric maximum-likelihood estimator

Tamer instead defines

$$H(x) = \Pr[(0, 1) \mid x]. \quad (25)$$

If $H(x)$ were known, one could use the richer likelihood

$$\begin{aligned} \mathcal{L}_{SML}(b; H) = \sum_i \left\{ 1[(0, 0)] \log P_1(x_i, b) + 1[(1, 1)] \log P_2(x_i, b) \right. \\ \left. + 1[(0, 1)] \log H(x_i) + 1[(1, 0)] \log (1 - P_1(x_i, b) - P_2(x_i, b) - H(x_i)) \right\}. \end{aligned} \quad (26)$$

Because $H(x)$ is unknown, the paper estimates it nonparametrically with a kernel estimator,

$$\hat{H}_n(x) = \frac{\sum_{i=1}^N 1[(Y_{1i}, Y_{2i}) = (0, 1)] K\left(\frac{x - x_i}{h}\right)}{\sum_{i=1}^N K\left(\frac{x - x_i}{h}\right)}, \quad (27)$$

and then trims it to respect the model bounds:

$$H_{tn}(x, b) = \min \left\{ P_4(x, b), \max [P_3(x, b), \hat{H}_n(x)] \right\}. \quad (28)$$

The SML estimator is

$$\hat{\beta}_{SML} = \arg \max_b \mathcal{L}_{SML}(b; H_{tn}). \quad (29)$$

Interpretation. The estimator uses the data to learn how often $(0, 1)$ occurs, but it never lets the nonparametric estimate violate the structural inequality restrictions of the game.

3.8 Large-sample result for the SML estimator

Appendix theorem 5 shows that, under kernel and regularity conditions, the SML estimator is consistent and $\sqrt{n}$ -normal:

$$\sqrt{n}(\hat{\beta}_{SML} - \beta) \Rightarrow N(0, Q_{SML}), \quad (30)$$

with

$$Q_{SML} = \left\{ E_x \left[\frac{\partial P_1}{P_1} \frac{\partial P_1'}{P_1} + \frac{\partial P_2}{P_2} \frac{\partial P_2'}{P_2} + \frac{(\partial P_1 + \partial P_2)(\partial P_1 + \partial P_2)'}{1 - P_1 - P_2 - H(x)} \right] \right\}^{-1}. \quad (31)$$

Interpretation. Relative to the limited ML estimator, the SML likelihood separates $(0, 1)$ from $(1, 0)$. That extra information appears exactly through the $H(x)$ term.

3.9 Theorem 3: efficiency gain with multiple equilibria

The paper's headline efficiency result is:

Theorem 3. If $\Pr[(0, 1) \mid x] > 0$, then the SML estimator is more efficient than the limited ML estimator.

The proof is simple but sharp:

$$1 - P_1 - P_2 - H < 1 - P_1 - P_2, \quad (32)$$

so the information matrix under SML is larger, which makes the asymptotic covariance matrix smaller.

Interpretation. Multiple equilibria create an opportunity to estimate $\Pr[(0, 1) \mid x]$ separately. Once that outcome is exploited rather than lumped together with $(1, 0)$, the parameter estimates become sharper.

4 What makes the results credible (assumptions, robustness, and limits)

4.1 Why the paper rejects the usual coherency fix

The standard coherency fix $\Delta_1 \Delta_2 = 0$ guarantees a unique reduced form, but it also makes the model recursive and removes the simultaneity the researcher may care about.

Interpretation. Tamer's argument is methodological: do not impose coherency just to obtain a familiar likelihood if doing so destroys the economics of strategic interaction.

4.2 What identifies the same-sign model

Identification is driven by:

- exact probabilities for some outcomes,
- excluded continuous variation,
- full-rank regressor matrices,
- and rich support that lets the researcher compare probabilities in extreme regions of the covariate space.

Interpretation. The paper is not identifying β from a single moment or local derivative. It identifies β from how the entire geometry of the outcome regions shifts with excluded regressors.

4.3 Why the opposite-sign case is harder but still identifiable

When $\Delta_1 \Delta_2 < 0$, there are no exact-probability outcomes to anchor the likelihood. The paper therefore works directly with lower and upper probability bounds and shows that, for a wrong parameter vector b , the lower bound under the truth can exceed the upper bound under b on a positive-probability set.

Interpretation. This is the key credibility point of theorem 2: even interval restrictions can be sharp enough to rule out all wrong parameter vectors.

4.4 Role of the distributional assumptions

The paper assumes a known bivariate shock distribution F_u up to the covariance matrix Ω , and independence between U and X . Tamer explicitly notes:

- known F_u is convenient for identification and likelihood methods but is stronger than strictly necessary for identification;
- the independence assumption could be weakened, but only at the cost of much stronger assumptions on $U \mid X$.

Interpretation. The paper is honest about where the structure comes from. Its identification results are not purely nonparametric, but they also do not rely on arbitrary equilibrium-selection assumptions.

4.5 Mixed-strategy equilibria

The main exposition rules out mixed strategies. If mixed strategies are allowed in the same-sign case, additional probability mass can appear in the “square” multiplicity region. The paper shows how the bounds adjust by adding a $P_{\square}(x, \beta)$ term.

Interpretation. The framework is flexible enough to accommodate mixed strategies, but the cleanest efficiency results come from the case where some outcomes remain exactly predicted.

4.6 What the paper can and cannot distinguish

The paper shows that the model is informative, but it does *not* fully solve equilibrium selection. In particular:

- it does not specify a structural rule that selects one equilibrium inside the multiplicity region;
- it estimates probabilities or bounds for outcomes in that region instead;
- and in the everywhere-incomplete case it points toward set-estimation or minimum-distance methods rather than a single exact likelihood.

Interpretation. The paper’s contribution is to show that the econometrician does not need to know the equilibrium-selection rule in order to identify and estimate the structural parameters in many important cases.

4.7 Relation to the literature

The paper sits at the intersection of several literatures:

- **Heckman (1978), Schmidt (1981), Gouriéroux–Laffont–Monfort:** the coherency literature in simultaneous discrete systems;
- **Bresnahan and Reiss, Berry:** empirical entry models that often simplify multiplicity away;
- **Bjorn and Vuong, Kooreman:** approaches that impose a known selection probability in the multiplicity region;
- **Manski and Tamer (2002):** incomplete models defined by inequality restrictions.

Interpretation. Tamer’s key advance is to replace the “avoid multiplicity” mindset with an “exploit multiplicity” mindset.

5 Figures and key theoretical results

5.1 Table 1: the basic simultaneous discrete-response game

TABLE 1

Discrete game with stochastic payoff

	$y_2 = 0$	$y_2 = 1$
$= 0$	$0, 0$	$0, x_2\beta_2 - u_2$
$= 1$	$x_1\beta_1 - u_1, 0$	$x_1\beta_1 + \Delta_1 - u_1, x_2\beta_2 + \Delta_2 - u_2$

Read:

- The 2×2 payoff table maps the binary game directly into the simultaneous latent-index system.
- The off-diagonal cells contain the interaction terms Δ_1 and Δ_2 , which determine whether strategic interaction produces complementarity or substitutability.
- The same table is the source of the multiplicity problem: for some values of u_1 and u_2 , more than one action profile satisfies best responses.

Economic message. The table makes clear that the econometric incompleteness is not an arbitrary statistical issue. It comes directly from an economically meaningful game.

5.2 Figure 1: multiplicity region when $\Delta_1, \Delta_2 < 0$

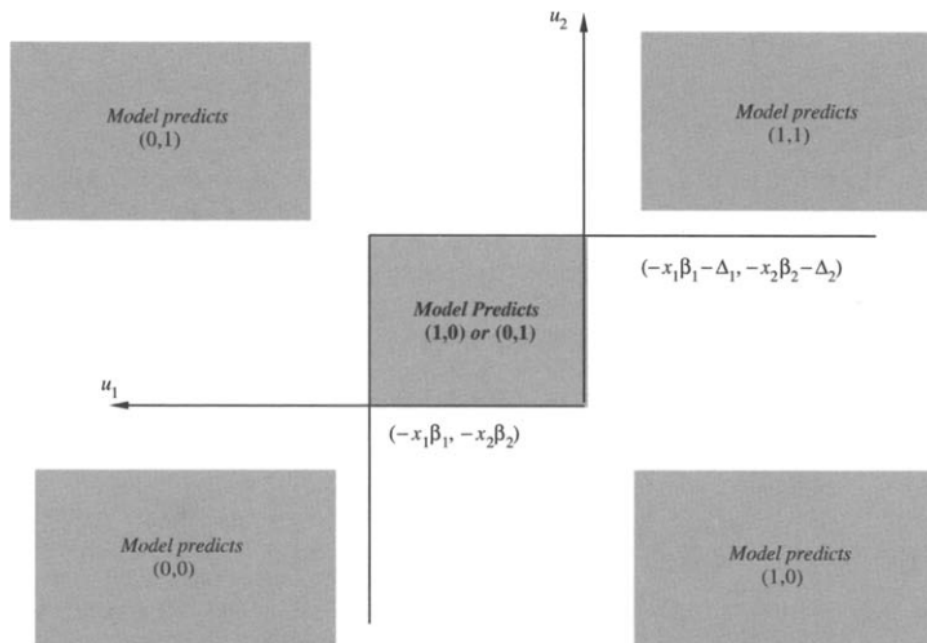


FIGURE 1
Incomplete model with multiple equilibria

Read:

- The four quadrants show where the model uniquely predicts (0,0), (1,1), (0,1), and (1,0).
- The shaded central box is the region where the model predicts either (1,0) or (0,1).
- That middle box is exactly why the econometric model is incomplete rather than coherent.

Economic message. For the same-sign case, the model is sharp on the diagonal outcomes and incomplete only in the middle. That structure is what lets Tamer get point identification and efficiency gains.

5.3 Table 2: Cournot entry example

TABLE 2
Payoffs in a Cournot model

	No entry	Enter
No entry	0, 0	0, $\frac{a^2}{4b} + x_2\beta_2 - u_2$
Enter	$\frac{a^2}{4b} + x_1\beta_1 - u_1, 0$	$\frac{a^2}{9b} + x_1\beta_1 - u_1, \frac{a^2}{9b} + x_2\beta_2 - u_2$

Read:

- The payoff matrix for the two-firm Cournot entry problem generates the same simultaneous discrete-response system as the baseline model.
- The asymmetric “enter / no entry” outcomes can both be equilibria.
- The size of the negative interaction term is $-5a^2/(36b)$, so business stealing creates strategic substitutability.

Economic message. The example connects the abstract econometric problem to a canonical empirical IO setting. It shows that multiplicity is exactly what one would expect in realistic entry games.

5.4 Figure 2: opposite-sign case and everywhere-bounded outcomes

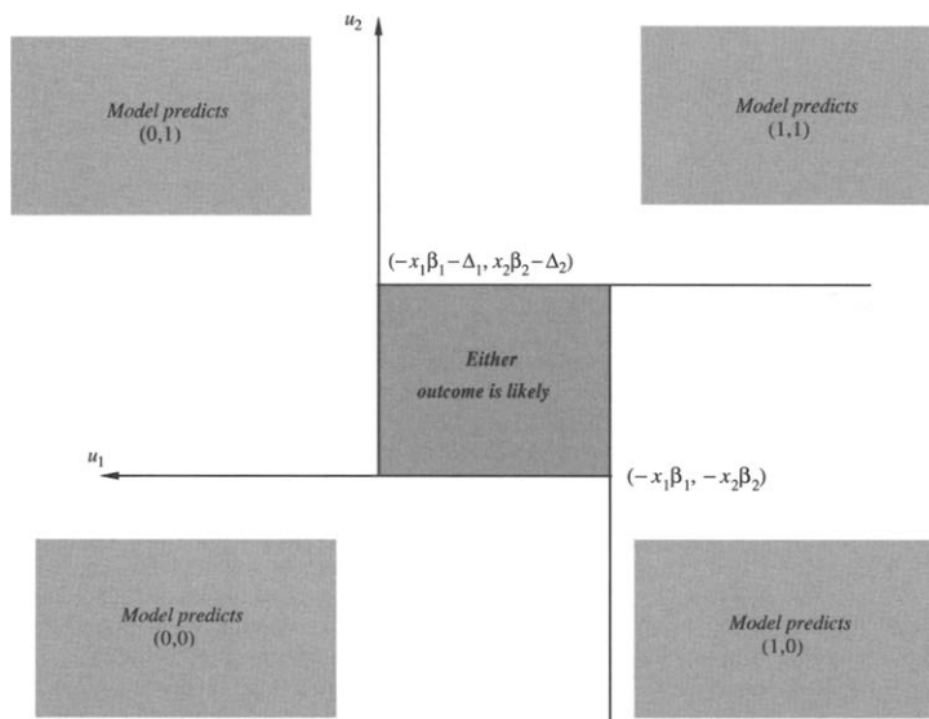


FIGURE 2
Incomplete bivariate model for the case where $\Delta_1 > 0$, $\Delta_2 < 0$

Read:

- The central square now says “Either outcome is likely,” reflecting the case where $\Delta_1 > 0$ and $\Delta_2 < 0$.
- Relative to figure 1, incompleteness is more severe: every outcome is bounded rather than some being known exactly.
- This is the case analyzed in theorem 2.

Economic message. The paper’s identification logic is strong enough to survive even when the model never delivers exact outcome probabilities pointwise in x .

5.5 Key result map

Result	What it says economically
Theorem 1	Same-sign interaction parameters do not prevent point identification; exact diagonal outcomes are enough to recover the full parameter vector.
Theorem 2	Even if all outcome probabilities are only bounded, excluded continuous variation can still point identify the model.
Theorem 3	Separating the two off-diagonal outcomes improves efficiency relative to collapsing them into one event.
Appendix theorem 5	The semiparametric likelihood estimator is consistent and $\sqrt{n}$ -normal after kernel estimation and trimming of the nonunique outcome probability.

Economic message. The paper turns multiplicity from a nuisance into a source of identifying and efficiency-improving information.

6 Short summary

- A simultaneous binary-response model generated by a two-player game can be incomplete because multiple equilibria create nonunique predicted outcomes.
- That incompleteness should not be confused with misspecification.
- When the interaction terms have the same sign, some outcomes are exactly predicted and others are bounded; this is enough for point identification under a continuous excluded regressor.
- When the interaction terms have opposite signs, all outcomes are bounded, but the full set of inequality restrictions can still identify the parameter vector.
- A semiparametric likelihood estimator that respects the probability bounds is consistent, asymptotically normal, and more efficient than estimators that collapse the multiplicity region into one pooled event.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that simultaneous discrete-response models with multiple equilibria do not need to be rescued by ad hoc equilibrium selection or by coherency restrictions that destroy simultaneity. Incomplete models can still deliver point identification and practical estimators.

7.2 Economic intuition

Multiplicity creates a region where the model predicts a set of feasible outcomes rather than a single one. That set still contains information: it restricts outcome probabilities, and those restrictions move in disciplined ways with the covariates. If the covariates have enough variation, the wrong parameter vectors cannot reproduce those restrictions.

7.3 Takeaway

Incomplete Simultaneous Discrete Response Model with Multiple Equilibria is a foundational paper because it changes how empirical researchers should think about multiplicity. Multiple equilibria are not necessarily an econometric dead end; with the right structure, they generate inequalities that can identify the model and improve efficiency when estimation uses them correctly.